```
clc;
clear;
close all;

% ========================================================
% DESIGN AND COMPARATIVE ANALYSIS OF IIR FILTERS
% Butterworth, Chebyshev Type-I and Elliptic
% ========================================================

% 1. INPUT PARAMETERS
Fs = 1000;       % Sampling frequency (Hz)
Fp = 100;        % Passband edge frequency (Hz)
Fst = 150;       % Stopband edge frequency (Hz)
Rp = 1;          % Passband ripple (dB)
Rs = 40;         % Stopband attenuation (dB)

% Normalized frequencies
Wp = Fp/(Fs/2);
Wst = Fst/(Fs/2);

% ========================================================
% 2. CHEBYSHEV TYPE-I THEORETICAL FORMULA
% ========================================================
epsilon = sqrt(10^(Rp/10)-1);

N_formula = acosh(sqrt((10^(Rs/10)-1)/ ...
    (10^(Rp/10)-1))) / acosh(Fst/Fp);

N_theory = ceil(N_formula);

% ========================================================
% 3. BUTTERWORTH FILTER
% ========================================================
[Nb, Wb] = buttord(Wp, Wst, Rp, Rs);
[bB, aB] = butter(Nb, Wb, 'low');

% ========================================================
% 4. CHEBYSHEV TYPE-I FILTER
% ========================================================
[Nc, Wc] = cheb1ord(Wp, Wst, Rp, Rs);
[bC, aC] = cheby1(Nc, Rp, Wc, 'low');

% ========================================================
% 5. ELLIPTIC FILTER
% ========================================================
[Ne, We] = ellipord(Wp, Wst, Rp, Rs);
[bE, aE] = ellip(Ne, Rp, Rs, We, 'low');

% ========================================================
% 6. DISPLAY INPUT PARAMETERS
% ========================================================
fprintf('\n=========================================\n');
```

```
fprintf(' IIR FILTER COMPARATIVE ANALYSIS\n');
fprintf('=========================================\n');

fprintf('\nINPUT PARAMETERS\n');
fprintf('Sampling Frequency = %.2f Hz\n', Fs);
fprintf('Passband Edge = %.2f Hz\n', Fp);
fprintf('Stopband Edge = %.2f Hz\n', Fst);
fprintf('Passband Ripple = %.2f dB\n', Rp);
fprintf('Stopband Attenuation = %.2f dB\n', Rs);

% ========================================================
% 7. CHEBYSHEV FORMULA RESULTS
% ========================================================
fprintf('\nCHEBYSHEV TYPE-I FORMULA\n');
fprintf('Ripple factor epsilon = %.4f\n', epsilon);
fprintf('Calculated order = %.4f\n', N_formula);
fprintf('Theoretical minimum order = %d\n', N_theory);

% ========================================================
% 8. FILTER ORDER COMPARISON
% ========================================================
fprintf('\nFILTER ORDER COMPARISON\n');
fprintf('Butterworth Order = %d\n', Nb);
fprintf('Chebyshev Type-I Order = %d\n', Nc);
fprintf('Elliptic Order = %d\n', Ne);

% ========================================================
% 9. DISPLAY FILTER COEFFICIENTS
% ========================================================
fprintf('\nBUTTERWORTH COEFFICIENTS\n');
fprintf('Numerator bB =\n');
disp(bB);
fprintf('Denominator aB =\n');
disp(aB);

fprintf('\nCHEBYSHEV TYPE-I COEFFICIENTS\n');
fprintf('Numerator bC =\n');
disp(bC);
fprintf('Denominator aC =\n');
disp(aC);

fprintf('\nELLIPTIC COEFFICIENTS\n');
fprintf('Numerator bE =\n');
disp(bE);
fprintf('Denominator aE =\n');
disp(aE);

% ========================================================
% 10. FREQUENCY RESPONSE
% ========================================================
N = 4096;

[HB, f] = freqz(bB, aB, N, Fs);
[HC, ~] = freqz(bC, aC, N, Fs);
```

```
[HE, ~] = freqz(bE, aE, N, Fs);

MB = 20*log10(max(abs(HB), 1e-12));
MC = 20*log10(max(abs(HC), 1e-12));
ME = 20*log10(max(abs(HE), 1e-12));

% ========================================================
% 11. PASSBAND RIPPLE
% ========================================================
P = f <= Fp;

Ripple_B = max(MB(P)) - min(MB(P));
Ripple_C = max(MC(P)) - min(MC(P));
Ripple_E = max(ME(P)) - min(ME(P));

% ========================================================
% 12. STOPBAND ATTENUATION
% ========================================================
S = f >= Fst;

Stop_B = -max(MB(S));
Stop_C = -max(MC(S));
Stop_E = -max(ME(S));

% ========================================================
% 13. PERCENTAGE DEVIATION
% ========================================================
Dev_C_Ripple = abs(Ripple_C - Rp)/Rp*100;

Dev_B_Stop = abs(Stop_B - Rs)/Rs*100;
Dev_C_Stop = abs(Stop_C - Rs)/Rs*100;
Dev_E_Stop = abs(Stop_E - Rs)/Rs*100;

% ========================================================
% 14. TRANSITION CHARACTERISTICS
% ========================================================
TransitionBW = Fst - Fp;

% ========================================================
% 15. DISPLAY FILTER PERFORMANCE
% ========================================================
fprintf('\n============================================\n');
fprintf(' FILTER PERFORMANCE\n');
fprintf('============================================\n');

fprintf('\nPASSBAND RIPPLE\n');
fprintf('Butterworth = %.4f dB\n', Ripple_B);
fprintf('Chebyshev Type-I = %.4f dB\n', Ripple_C);
fprintf('Elliptic = %.4f dB\n', Ripple_E);

fprintf('\nSTOPBAND ATTENUATION\n');
fprintf('Butterworth = %.4f dB\n', Stop_B);
fprintf('Chebyshev Type-I = %.4f dB\n', Stop_C);
fprintf('Elliptic = %.4f dB\n', Stop_E);
```

```
fprintf('\nCHEBYSHEV RIPPLE DEVIATION\n');
fprintf('Specified Ripple = %.2f dB\n', Rp);
fprintf('Measured Ripple = %.4f dB\n', Ripple_C);
fprintf('Percentage Deviation = %.4f %%\n', Dev_C_Ripple);

fprintf('\nSTOPBAND ATTENUATION DEVIATION\n');
fprintf('Butterworth = %.4f %%\n', Dev_B_Stop);
fprintf('Chebyshev Type-I = %.4f %%\n', Dev_C_Stop);
fprintf('Elliptic = %.4f %%\n', Dev_E_Stop);

fprintf('\nTRANSITION CHARACTERISTICS\n');
fprintf('Passband Edge = %.2f Hz\n', Fp);
fprintf('Stopband Edge = %.2f Hz\n', Fst);
fprintf('Transition Bandwidth = %.2f Hz\n', TransitionBW);

% ========================================================
% 16. MAGNITUDE RESPONSE
% ========================================================
figure('Name', 'IIR Magnitude Response', 'NumberTitle', 'off');

plot(f, MB, 'LineWidth', 1.5);
hold on;
plot(f, MC, 'LineWidth', 1.5);
plot(f, ME, 'LineWidth', 1.5);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Comparison of IIR Filter Magnitude Responses');

legend('Butterworth', 'Chebyshev Type-I', 'Elliptic', ...
    'Location', 'best');

xlim([0 250]);
ylim([-80 5]);

% Reference lines
xline(Fp, '--', 'Fp = 100 Hz');
xline(Fst, '--', 'Fst = 150 Hz');
yline(-Rp, '--', 'Rp = -1 dB');
yline(-Rs, '--', 'Rs = -40 dB');

hold off;

% ========================================================
% 17. PHASE RESPONSE
% ========================================================
Phase_B = unwrap(angle(HB))*180/pi;
Phase_C = unwrap(angle(HC))*180/pi;
Phase_E = unwrap(angle(HE))*180/pi;

figure('Name', 'IIR Phase Response', 'NumberTitle', 'off');
```

```
plot(f, Phase_B, 'LineWidth', 1.2);
hold on;
plot(f, Phase_C, 'LineWidth', 1.2);
plot(f, Phase_E, 'LineWidth', 1.2);

grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Comparison of IIR Filter Phase Responses');

legend('Butterworth', 'Chebyshev Type-I', 'Elliptic', ...
    'Location', 'best');

xlim([0 250]);
hold off;

% ========================================================
% 18. FILTER ORDER GRAPH
% ========================================================
figure('Name', 'Filter Order Comparison', 'NumberTitle', 'off');

bar([Nb Nc Ne]);

set(gca, 'XTick', 1:3);
set(gca, 'XTickLabel', ...
    {'Butterworth', 'Chebyshev-I', 'Elliptic'});

ylabel('Filter Order');
xlabel('Filter Type');
title('Comparison of IIR Filter Orders');

grid on;


==========================================
 IIR FILTER COMPARATIVE ANALYSIS
==========================================

INPUT PARAMETERS
Sampling Frequency = 1000.00 Hz
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Passband Ripple = 1.00 dB
Stopband Attenuation = 40.00 dB

CHEBYSHEV TYPE-I FORMULA
Ripple factor epsilon = 0.5088
Calculated order = 6.2071
Theoretical minimum order = 7

FILTER ORDER COMPARISON
Butterworth Order = 12
Chebyshev Type-I Order = 6
Elliptic Order = 4
```

```
BUTTERWORTH COEFFICIENTS
Numerator bB =
   1.0e-03 *

  Columns 1 through 7

    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050

  Columns 8 through 13

    0.1757    0.1098    0.0488    0.0146    0.0027    0.0002

Denominator aB =
  Columns 1 through 7

    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643

  Columns 8 through 13

  -27.6092   11.5218   -3.4662    0.7128   -0.0899    0.0053


CHEBYSHEV TYPE-I COEFFICIENTS
Numerator bC =
   1.0e-03 *

    0.0464    0.2782    0.6956    0.9274    0.6956    0.2782    0.0464

Denominator aC =
    1.0000   -4.8694   10.3807  -12.3367    8.5969   -3.3268    0.5586


ELLIPTIC COEFFICIENTS
Numerator bE =
    0.0197   -0.0171    0.0333   -0.0171    0.0197

Denominator aE =
    1.0000   -3.0330    3.8118   -2.2911    0.5554


==========================================
 FILTER PERFORMANCE
==========================================

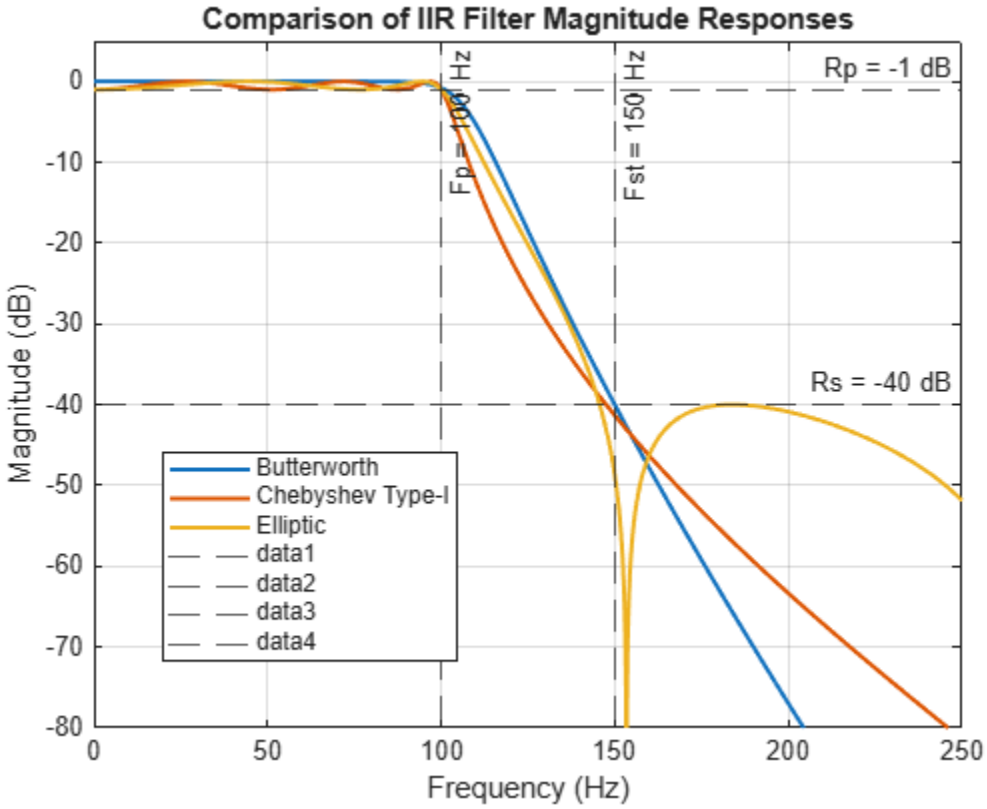
PASSBAND RIPPLE
Butterworth = 0.8033 dB
Chebyshev Type-I = 1.0000 dB
Elliptic = 1.0000 dB

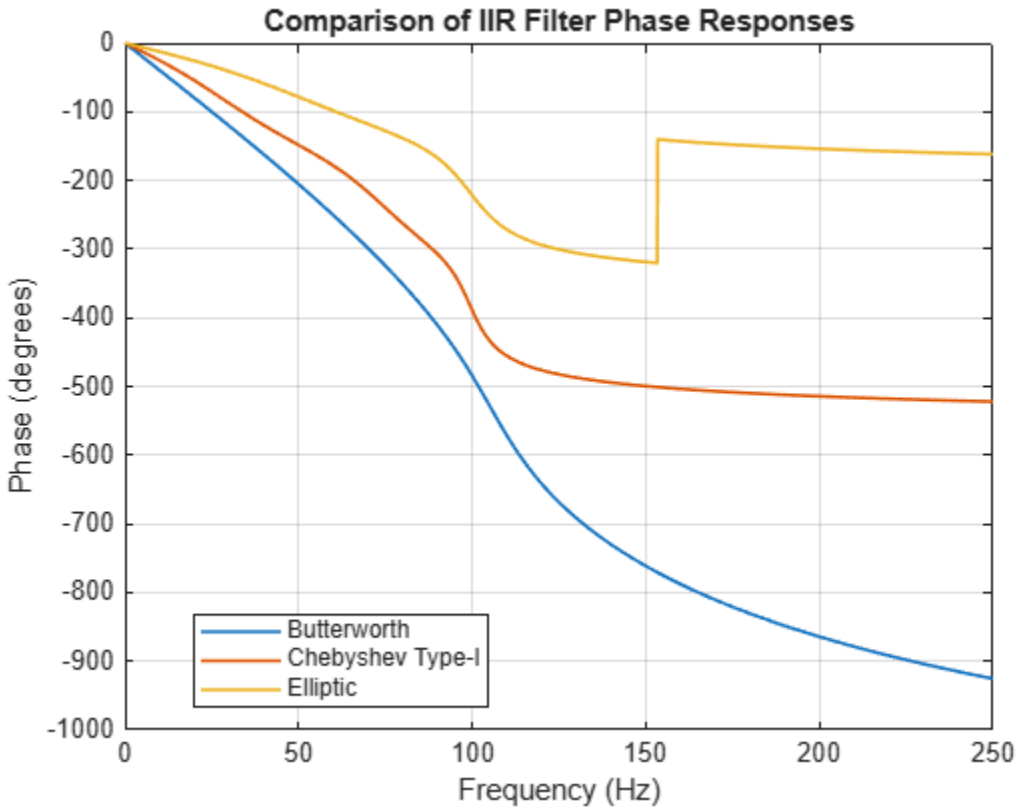
STOPBAND ATTENUATION
Butterworth = 40.0198 dB
Chebyshev Type-I = 41.3365 dB
Elliptic = 40.0000 dB
```

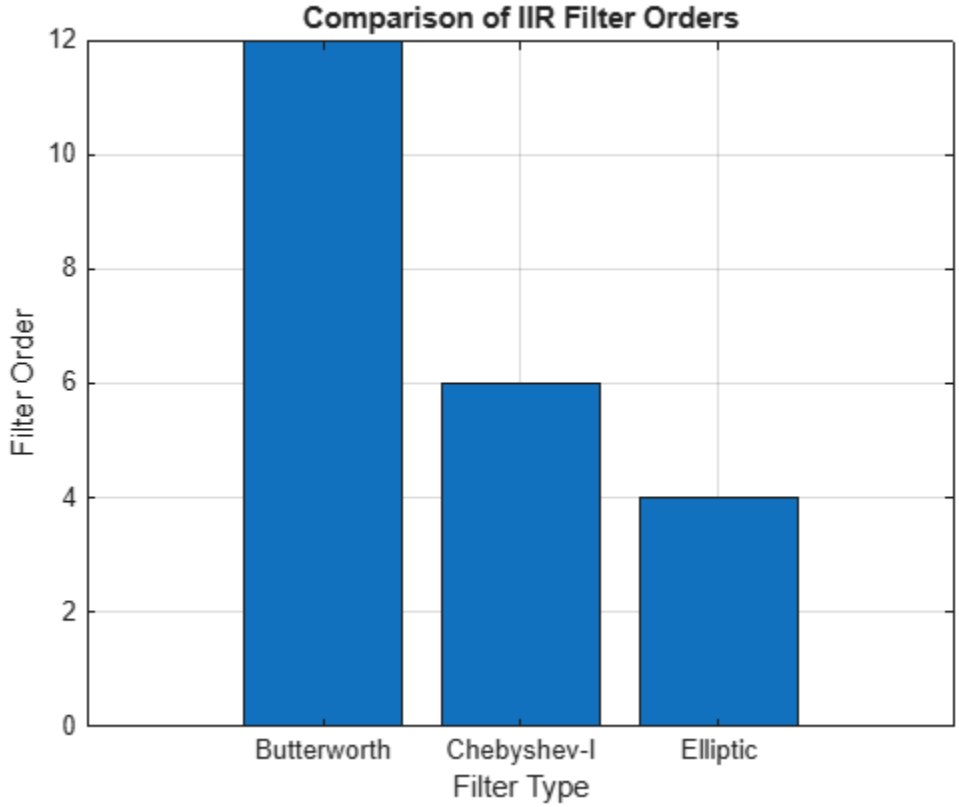
CHEBYSHEV RIPPLE DEVIATION
Specified Ripple = 1.00 dB
Measured Ripple = 1.0000 dB
Percentage Deviation = 0.0000 %

STOPBAND ATTENUATION DEVIATION
Butterworth = 0.0494 %
Chebyshev Type-I = 3.3413 %
Elliptic = 0.0000 %

TRANSITION CHARACTERISTICS
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Transition Bandwidth = 50.00 Hz




Comparison of IIR Filter Phase Responses
Phase (degrees)
Frequency (Hz)
0
-100
-200
-300
-400
-500
-600
-700
-800
-900
-1000
0
50
100
150
200
250
Butterworth
Chebyshev Type-I
Elliptic


Comparison of IIR Filter Orders
Filter Order
Filter Type
12
10
8
6
4
2
0
Butterworth
Chebyshev-I
Elliptic

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